

Alexander Biddle

Linked-In: www.linkedin.com/in/alex-biddle12

GitHub: <https://github.com/AlexanderBiddle>

Contact: Alexbid2004@gmail.com

Professional Summary

Security-focused Cybersecurity and Computer Science student with strong foundations in systems programming, secure software development, and networked applications. Experienced in building and analyzing secure client-server systems, applying SSDLC principles, and performing threat-driven security analysis in mission-oriented environments. Proven ability to work across low-level systems, networking, and security domains, with demonstrated leadership, teamwork, and communication skills through internships and collegiate athletics. Seeking opportunities to contribute to security-critical systems in defense, government, or mission-driven environments.

Education - Randolph-Macon College (Dean's List • ODAC All-Academic Award • ODAC All-Conference Award)

Major(s): Cybersecurity & Computer Science

CS & Cybersecurity GPA: 4.0

Minor(s): Engineering Physics & Mathematics

Technical Skills and Abilities

Languages: Python, C, C++, Rust, Java, JavaScript, MIPS

Tools & Environments: Windows, Linux, SSH, Apache, MATLAB, AFSIM, Git, Overleaf, Wireshark, nmap, tcpdump

Certifications: CompTIA Security+, Network+, Cloud essentials

- Designing and implementing secure, object-oriented software using efficient data structures and algorithms, with emphasis on correctness, maintainability, and defensive design
- Applying secure coding practices with strict input validation, access control, memory-safe programming, and mitigation of common vulnerabilities (buffer overflows, race conditions, injection, insecure deserialization)
- Implementing layered security controls across multi-level systems and applications, including authentication, authorization, encryption, input validation, and audit logging
- Systems-level programming in C/C++ with focus on memory management, pointers, process control, and concurrency
- Conducted threat-driven security analysis of systems and networks by modeling attacker behavior, identifying attack surfaces, and evaluating failure modes in mission-critical environments with documentation and analysis
- Implemented and analyzed Denial-of-Service (DoS) attacks including UDP floods, ICMP floods, and ICMP reflection attacks, measuring resource exhaustion, packet loss, latency degradation, and service availability impacts
- Evaluated network behavior under attack using packet capture, traffic generation, and system monitoring tools (*e.g., hping3, tcpdump, Wireshark, nmap, bmon, ping*), correlating traffic volume, payload size, and system responsiveness.
- Applied System Security & Defense principles to assess system hardening effectiveness, protocol misuse, spoofed source addressing, and resilience against network-based attacks
- Implemented and evaluated secure client-server communication paths using TCP/IP, UDP, and HTTP/HTTPS, emphasizing confidentiality, integrity, authentication, session management, and encrypted data exchange
- Demonstrated strong understanding of network-level security concepts, including protocol trust boundaries, attack surfaces, spoofing risks, reflection/amplification attacks, and defense strategies across OSI and Internet stack layers 1

Experiences

SSEP Mission Analyst Intern

June 2024 – Present

Naval Surface Warfare Center Dahlgren Division, M Department

- Developed simulation models in Advanced Framework Simulation and Integration Modeling (AFSIM) tool to run simulated scenarios in support of mission analysis.
- Developed behavior logic and interfaces to model system responses to threat conditions for Red vs Blue scenarios in AFSIM
- Supported collaborative projects in completion of strategy concepts and task management.
- Created analysis scripts in MATLAB to process MONTE CARLO outputs of AFSIM

IT Intern

December 2023 – January 2024

C2 – Essentials

- Provided technical support to internal staff, assisted senior developers with website content management and system maintenance
- Conducted maintenance, performance optimization and addressing computer-related issues, and troubleshooting hardware and software problems.

Projects

CipherSafe — Zero-Knowledge Cryptographic Vault Platform		
▪ Designed and implemented a zero-knowledge password vault in which all sensitive fields, including vault metadata, remain encrypted end-to-end, preventing server-side plaintext exposure even under database compromise ▪ Engineered a custom four-way handshake protocol layered over TLS 1.3 and HTTPS 3.0 to enforce explicit cryptographic state transitions, session binding, replay resistance, and authenticated key exchange ▪ Implemented RSA-2048 OAEP for secure session key wrapping and RSA-PSS for digital signatures, enforcing cryptographic authenticity and integrity across all handshake phases ▪ Built AES-256-GCM session encryption with strict nonce tracking and replay detection, including per-session nonce sets, TTL-based expiration, and atomic session key rotation ▪ Designed a dual-salt Argon2id password derivation model separating client-side and server-side salts to harden against offline cracking, credential reuse, and rainbow-table attacks ▪ Enforced strict protocol-level validation to prevent base64 smuggling, oversized payloads, malformed ciphertext, and deserialization abuse, rejecting invalid input at every cryptographic boundary ▪ Implemented cryptographic key lifecycle management, including RSA key rotation with key identifiers, expiration enforcement, and backward-compatible handling of active sessions ▪ Built a centralized error normalization and audit layer that prevents cryptographic oracle leaks, timing disclosures, and sensitive error propagation while preserving security visibility		
Technology: Python, Flask, PostgreSQL, JavaScript TLS 1.3, HTTPS 3.0, RSA-OAEP/PSS, AES-256-GCM, Argon2id		
Security & Defense Labs		
▪ Conducted hands-on adversarial and defensive security analysis of systems and networks, executing controlled attacks to evaluate system behavior, resilience, and service availability ▪ Modeled network and system attack surfaces, identified protocol and configuration weaknesses, and assessed vulnerability impact under realistic threat scenarios ▪ Analyzed resource exhaustion, packet loss, and latency degradation during network-based attacks, correlating traffic characteristics with system performance ▪ Implemented and evaluated mitigation considerations by reasoning about system hardening, protocol misuse prevention, and defensive controls based on observed attack outcomes		
Collegiate Courses		
▪ CS & Cyber: Data Structures, Systems Programming, Computer Organization, Object-Oriented Programming, Algorithms, Machine Learning, Privacy and Security, Software Security, Networks Architecture and Protocols, Computer and Data Security, Operating Systems, System Security & Defense ▪ Math & Engineering: Calculus I & II, Differential Equations, Multivariable Calculus, Higher Geometry, Linear Algebra, Mathematical Physics, Physics I & II, Engineering Statics, Mechanics of Solids, Dynamics of Engineering Mechanics, Modern Physics		
Service		
Randolph Macon Football	August 2022 – January 2026	
Macon A Difference Day – Volunteer	Spring 2023, 2024, 2025	
Rise Against Hunger – Volunteer	Spring 2024, 2025	